

Ruihuan Cheng

Doctor of Philosophy

Born in 1995, Shandong province, China

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Research Interests

Theoretical Study on Material Physics

1. Machine learning potential and molecular dynamics
2. Lattice dynamics and thermal transport
3. Polyamide-based membranes
4. Superionic phase transition and composite systems

Education and Experience

- Oct. 2025 – Present **The University of Hong Kong** – Hong Kong SAR, China
Research Associate, Department of Mechanical Engineering
- Sept. 2021 – Sept. 2025 **The University of Hong Kong** – Hong Kong SAR, China
Ph.D. in Materials Physics, Department of Mechanical Engineering
Supervisor: Prof. Yue Chen
- Sept. 2018 – Jun. 2021 **Huazhong University of Science and Technology / MINES Paris Tech** – Wuhan, China / Paris, France
MEng in New Energy Science and Engineering / Renewable Energy Science and Technology, China-EU Institute for Clean and Renewable Energy
Supervisor: Prof. Song Li / Prof. Alain THOREL
- Sept. 2014 – Jun. 2018 **China University of Petroleum (East China)** – Qingdao, China
BEng in Petroleum Engineering, Institute of Petroleum Engineering
GPA: 88.42 (Ranking: 9/313)

Research Experience

- Jun. 2023 – Present **Shaping Gas Fuel Storage; Understanding the Thermal Properties of Porous Metal-Organic Frameworks**
Supervisor: Prof. Yue Chen.
Molecular dynamics simulations using a general force field were performed to study the thermal transport of MOFs in the gas adsorption state. To understand the relationship between surface functionalization, interfacial heat/mass transport and kinetic properties.
- Jan. 2023 – Present **Phonon and Diffusion Transport in ZrBeSi-type Zintl Compounds: Effects of Vacancy Ordering and High-Order Lattice Anharmonicity**
Supervisor: Prof. Yue Chen.
Constructing the state-of-the-art machine-learning interatomic potential with first principles-based accuracy to investigate vacancy ordering influences phonon and diffusion transport and the impact of high-order anharmonicity on phonon scattering, which reduces thermal conductivity at high temperatures.

- Jan. 2022 – Present **Population and Coherence Two-Channel Lattice Thermal Transport in Crystals with Strong Anharmonicity**
 Supervisor: Prof. Yue Chen.
 To reveal the mechanisms of four-phonon scattering beyond the lowest order of perturbation theory, the population and coherence in lattice thermal transport are described more accurately using a two-channel model.
- Jan. 2020 – Jun. 2023 **Room-Temperature Hydrogen Storage Performance of Metal-Organic Framework (MOFs) Composites**
 Supervisor: Prof. Song Li.
 By combining MOFs with various nanomaterials, the research aims to enhance the hydrogen adsorption capacity, stability, and kinetics of MOF-based storage systems.
- May 2020 – Sept. 2020 **Simulation Study on Confined Growth of Polyamide-based Membrane**
 Supervisor: Prof. Song Li.
 Using molecular dynamics simulations to explore the confined growth of polyamide-based membranes. It investigates how confinement during membrane formation affects polymer structure, chain orientation, and interfacial interactions.
- Oct. 2019 – Oct. 2020 **Theoretical Study on Thermal Transport of Zeolitic Imidazolate Frameworks (ZIFs)**
 Supervisor: Prof. Song Li.
 Adopting molecular dynamics simulations to investigate the thermal conductivity of zeolitic imidazolate frameworks (ZIFs). Two new parameters, the alignment tensor (A_i) and the pathway factor (P_f), are proposed to evaluate the heat transfer pathways.

First author Publications

Co-first author, * Corresponding author

- 2025 **Strong Crystalline Thermal Insulation Induced by Extended Antibonding States**
Ruihuan Cheng, Chen Wang, Niuchang Ouyang, Xingchen Shen*, and Yue Chen*
Nat. Commun., 16(1): 7941.
- 2025 **Atomic Hopping Induced Dynamic Disorder Phonon Scattering and Suppressed Thermal Transport in Cu_4TiSe_4**
Ruihuan Cheng, Wenxuan Wang, Chen Wang, Siu Ting Tai, Niuchang Ouyang, and Yue Chen*
Newton, 1(4): 100090.
- 2024 **Impact of Strain-Insensitive Low-Frequency Phonon Modes on Lattice Thermal Transport in A_2XB_6 -type Perovskites**
Ruihuan Cheng, Zezhu Zeng, Chen Wang, Niuchang Ouyang, and Yue Chen*
Phys. Rev. B, 109 (5): 054305.
- 2023 **Lattice Dynamics and Thermal Transport of PbTe under High Pressure**
Ruihuan Cheng, Xingchen Shen, Stefan Klotz, Zezhu Zeng, Zehua Li, Alexandre Ivanov, Yu Xiao, Li-Dong Zhao, Frank Weber*, and Yue Chen*
Phys. Rev. B, 108 (10): 104306.
- 2023 **Understanding the Heat Transfer Performance of Zeolitic Imidazolate Frameworks upon Gas Adsorption by Molecular Dynamics Simulations**
Ruihuan Cheng, Wei Wei, Jincheng Zhang, and Song Li*
J. Phys. Chem. B, 127 (43): 9390-9398.
- 2022 **Polyamide-based Membranes with Structural Homogeneity for Ultrafast Molecular Sieving**
 Liang Shen[#], **Ruihuan Cheng**[#], Ming Yi[#], Wei-Song Hung, Susilo Japip, Lian Tian, Xuan Zhang, Shudong Jiang, Song Li*, and Yan Wang*
Nat. Commun., 13 (1): 500.
- 2021 **Molecular Insights into the Correlation between Microstructure and Thermal Conductivity of Zeolitic Imidazolate Frameworks**
Ruihuan Cheng, Wei Li, Wei Wei, Jun Huang, and Song Li*
ACS Appl. Mater. Interfaces, 13 (12): 14141-14149.

Co-author Publications

- 2025 **Revisit Many-body Interaction Heat Current and Thermal Conductivity Calculation using the Moment Tensor Potential LAMMPS Interface**
Siu Ting Tai, Chen Wang, **Ruihuan Cheng**, and Yue Chen*
J. Chem. Theory Comput., 21 (7): 3649-3657.
- 2025 **Positive Temperature-Dependent Thermal Conductivity induced by Wave-like Phonons in Complex Ag-based Argyrodites**
Niuchang Ouyang[#], Dongyi Shen[#], Chen Wang[#], **Ruihuan Cheng**, and Yue Chen*
Phys. Rev. B, 111 (6): 064307.
- 2024 **Pushing Thermal Conductivity to its Lower Limit in Crystals with Simple Structures**
Zezhu Zeng[#], Xingchen Shen[#], **Ruihuan Cheng**, Olivier Perez, Niuchang Ouyang, Zheyong Fan, Pierric Lemoine, Bernard Raveau, Emmanuel Guilmeau*, and Yue Chen*
Nat. Commun., 15 (1): 3007.
- 2024 **Switching from Negative to Positive Thermal Expansion of Porphyrin-based Metal-Organic Frameworks through Post-Metallization**
Zhanning Liu*, **Ruihuan Cheng**, Rui Ma, Chengyong Xing, Jian Tian, Yue Chen, and Xianran Xing*
Adv. Funct. Mater., 35(3): 2413290.
- 2024 **Enhanced Thermoelectric Performance and Mechanical Strength in GeTe Enable Power Generation and Cooling**
Jianglong Zhu, Fujie Zhang, Yilin Tai, Xiaobo Tan, Qian Deng, Pengfei Nan*, **Ruihuan Cheng**, Chengliang Xia, Yue Chen, Binghui Ge, and Ran Ang*
InfoMat, 6 (4): e12514.
- 2024 **Functionally Separated Electronic Band Engineering via Multi-Element Doping plus High-Density Defects Advances Broad-Temperature-Range Thermoelectric Performance in GeTe**
Jianglong Zhu, Xiaobo Tan, Dong Pan, Yanni Luo, Ruiheng Li, Xuri Rao, **Ruihuan Cheng**, Chengliang Xia, Yue Chen, Qiang Sun*, and Ran Ang*
Chem. Eng. J., 480: 148135.
- 2024 **Effects of Local Chemical Ordering on the Thermal Transport in Entropy-Regulated PbSe-based Thermoelectric Materials**
Shuang Lyu, **Ruihuan Cheng**, Haiqi Li, and Yue Chen*
Appl. Phys. Lett., 124 (23): 232202.
- 2024 **Molecular Dynamics Simulation Study on the Thermal Conductivity of Zeolitic Imidazolate Framework/*n*-Eicosane Composite Phase Change Materials**
Zijian Tan, Teng Wu, Yajun Qiao, **Ruihuan Cheng**, Wei Li*, Weixiong Wu
Acta Chim. Sinica, 82(12): 1193-1201.
- 2023 **Enhanced Thermoelectric Performance of *p*-type Bi₂Si₂Te₆ Enabled via Synergistically Optimizing Carrier Concentration and Suppressing Bipolar Effect**
Dongyi Shen, **Ruihuan Cheng**, Wenxuan Wang, Haiqi Li, Chen Chen, Qian Zhang, and Yue Chen*
Mater. Today Phys., 37: 101185.
- 2023 **Theoretical Evaluation of the Persistence of Transverse Phonons across a Liquid-like Transition in Superionic Conductor KAg₃Se₂**
Chen Wang, **Ruihuan Cheng**, and Yue Chen*
Chem. Mater., 35 (4): 1780-1787.
- 2023 **Ammonia Adsorption Performance of Zeolitic Imidazolate Frameworks for Cooling**
Zhilu Liu, **Ruihuan Cheng**, Juyeong Kim, and Song Li*
Langmuir, 39 (41): 14726-14736.

- 2023 **Understanding the Influence of Secondary Building Units on the Thermal Conductivity of Metal-Organic Frameworks via High-Throughput Computational Screening**
Yuanchuang Lin, **Ruihuan Cheng**, Tiangui Liang, Weixiong Wu, Song Li*, and Wei Li*
Phys. Chem. Chem. Phys., 25 (47): 32407-32415.

Presentations

- Jun. 2024 Strong Phonon Scattering and Suppressed Thermal Transport in Cu_4TiSe_4
The joint 40th International and 20th European Thermoelectric Conference, Krakow, Poland
- Aug. 2023 Lattice Dynamics and Thermal Transport of PbTe under High Pressure
Croucher Summer Course on Neutron Scattering, Hong Kong SAR, China
- Mar. 2023 Anomalous Pressure Dependence of Lattice Dynamics in PbTe: A Simulation Study
APS March Meeting 2023, Las Vegas, USA
- Nov. 2020 Molecular simulation study on the structure of zeolitic imidazolate framework and its correlation with thermal conductivity
2020 China Engineering Thermophysics Annual Conference on Heat and Mass Transfer, Guangzhou, China

Teaching experience

- Sept. 2022 – **The University of Hong Kong** – Hong Kong SAR, China
Sept. 2025 Teaching assistant for MECH3407 Advanced Partial Differential Equations and Complex Variables
- Feb. 2024 – **The University of Hong Kong** – Hong Kong SAR, China
Mar. 2024 Certificate in Teaching and Learning in Higher Education